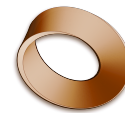


Name: \_\_\_\_\_

Date: \_\_\_\_\_

Mr. Rawson • Y8 MYP Physics



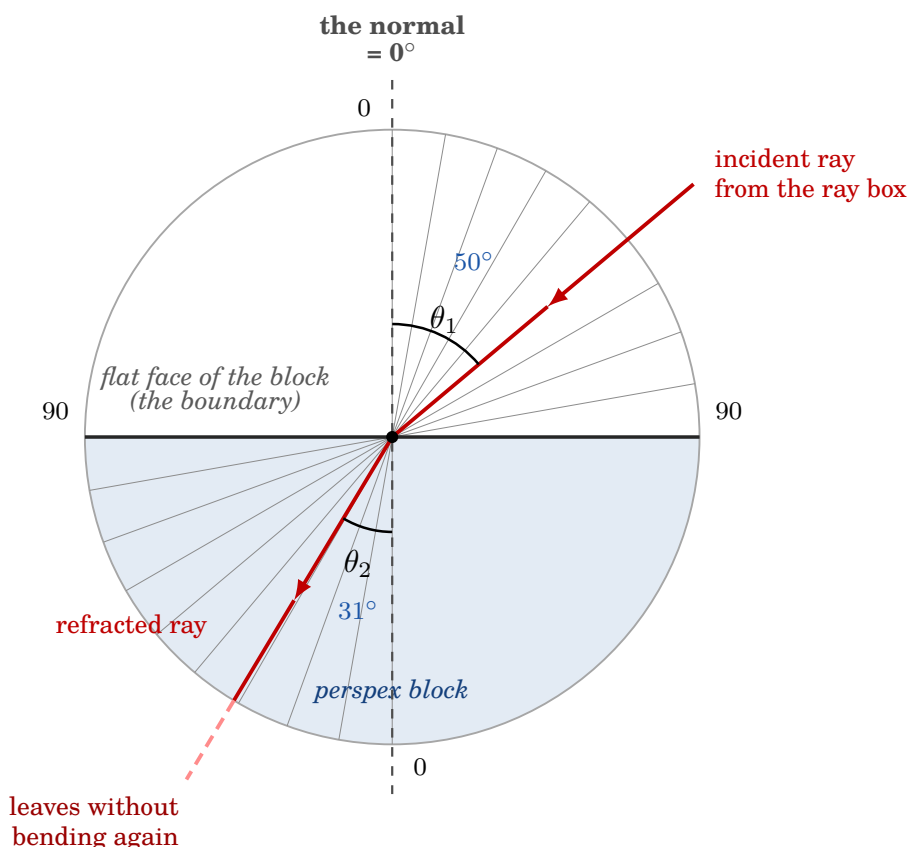
## 1.5 – Measuring Refraction

*Unit 1: Light & EM Spectrum*

### Learning intentions

Last lesson we marched the boundary ourselves and found out **why** light bends. Today we find out **how much** it bends — and we learn the way of measuring it that you will need for your investigation.

Your block is a **semicircular** piece of perspex, and it sits on a printed card with **two protractors** on it. The block goes in the blue half, with its **flat face** along the line across the middle. You aim the ray at the **centre** of the card — and then you read both angles straight off the card. No tracing. No lifting.



Three things to notice on that diagram before you touch any apparatus.

- The ray bends when it crosses the **flat face** — that is the boundary between air and perspex. It bends **towards the normal**, because perspex slows light down.

- **The normal is the dashed vertical line, and on this card it is 0°.** Both angles are measured from it.

- On the way out, the ray leaves through the **curved** face — and it does **not** bend again. Question 3(e) asks you why.

## 1. The method

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You have a **ray box**, a **semicircular perspex block**, and the **printed protractor card**. You will not need paper, a pencil trace, or your own protractor — the card is the protractor.

1. Put the block on the card in the **blue half**, curved side down. Line its **flat face** up exactly along the line across the middle, with the middle of the flat face on the **centre** of the card.
2. Switch the ray box on and make a **single** narrow ray. Aim it so that it hits the **centre** of the card — the point where the two protractors meet. Check this **every time**, before every reading.
3. Swing the ray box round until the incident ray lies along the  $0^\circ$  line of the top protractor. That is your first reading: an angle of incidence of  $0^\circ$ .
4. Read where the refracted ray comes out on the **bottom** protractor. Write it in the table.
5. Swing the ray box round to  $10^\circ$ , then  $20^\circ$ , and so on up to  $80^\circ$ , reading the bottom protractor each time.

### The two instructions printed on your card

“Set up your ray box so that the light ray hits the **centre** of the protractors every time you take a reading.”

“Remember to measure from the **normal** ( $90^\circ$  from the surface it is hitting), and take **the normal as  $0^\circ$** .”

### Finding the right scale — do this first

Your card has **two** scales printed on it. The one you want is the one that reads  **$0^\circ$  at the dashed normal**, counting up to  $90^\circ$  along the flat face.

**Here is the check.** Do step 3 — send the ray straight down the normal. **The scale you want reads  $0^\circ$** . Find that scale once, at the start, and every reading you take afterwards is measured from the right place.

**Safety.** The ray box gets **hot**. Do not touch the bulb, and keep paper, sleeves and pencil cases out of it. If you smell burning, switch it off at the power supply and tell Mr. Rawson.

## 2. Your results

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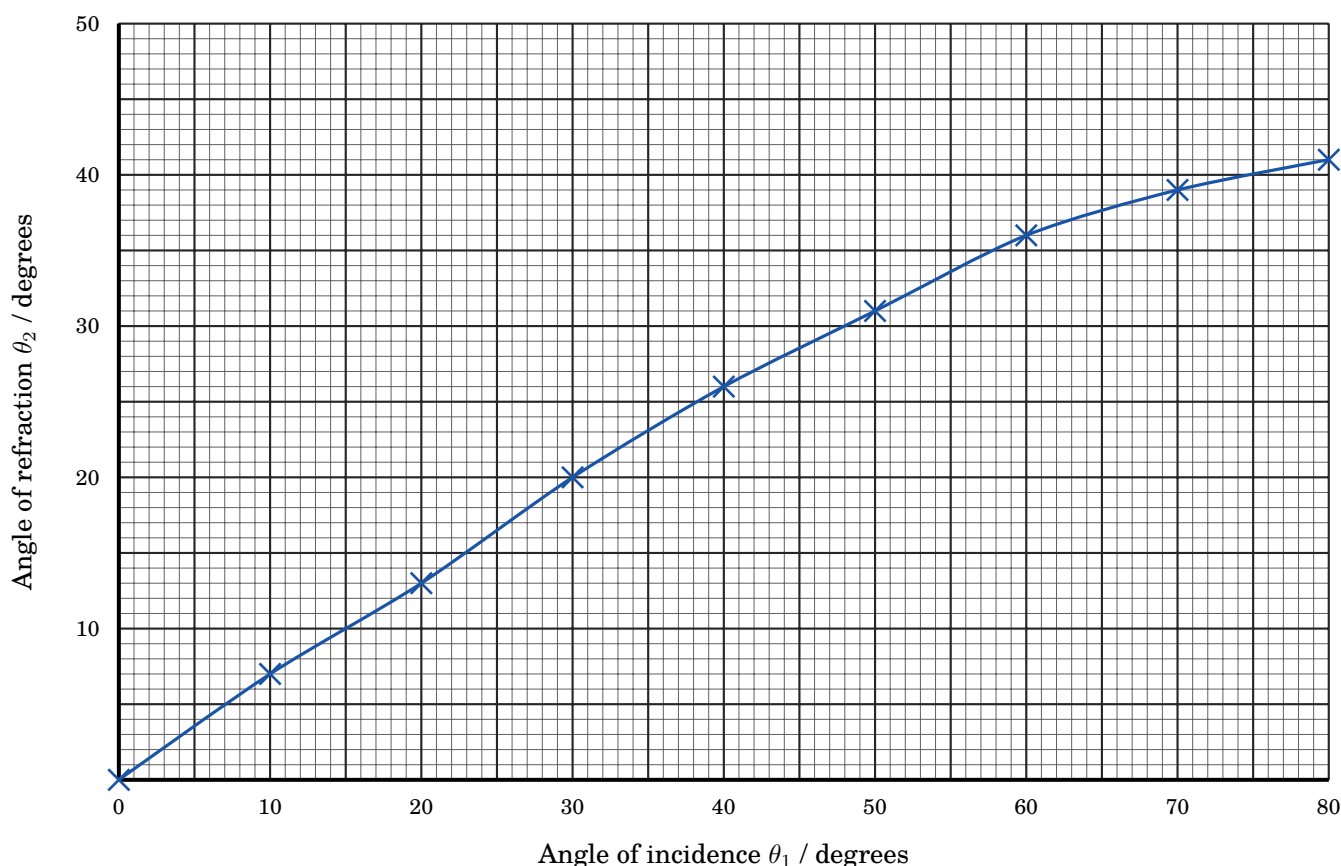
Read the angle of refraction to the **nearest degree**. The third column is just the first two subtracted — it tells you *how much* the ray bent.

| Angle of incidence<br>$\theta_1$ , in air | Angle of refraction<br>$\theta_2$ , in perspex | How much it bent<br>$\theta_1 - \theta_2$ |
|-------------------------------------------|------------------------------------------------|-------------------------------------------|
| $0^\circ$                                 | $0^\circ$                                      | $0^\circ$                                 |
| $10^\circ$                                | $\approx 7^\circ$                              | $3^\circ$                                 |
| $20^\circ$                                | $\approx 13^\circ$                             | $7^\circ$                                 |
| $30^\circ$                                | $\approx 20^\circ$                             | $10^\circ$                                |
| $40^\circ$                                | $\approx 26^\circ$                             | $14^\circ$                                |
| $50^\circ$                                | $\approx 31^\circ$                             | $19^\circ$                                |
| $60^\circ$                                | $\approx 36^\circ$                             | $24^\circ$                                |
| $70^\circ$                                | $\approx 39^\circ$                             | $31^\circ$                                |
| $80^\circ$                                | $\approx 41^\circ$                             | $39^\circ$                                |

Values for perspex,  $n = 1.49$ . Anything within two or three degrees is a good measurement. What matters is that  $\theta_2 < \theta_1$  every time, that the  $0^\circ$  row really does read  $0^\circ$ , and that the third column **grows** — the ray bends more and more as  $\theta_1$  increases.

### 3. Draw the graph

Plot **angle of refraction** (up) against **angle of incidence** (across). Plot your nine points with a sharp pencil, then join them with one smooth curve.



*The curve is the giveaway. It rises steeply at first and then **flattens off** — it is not a straight line, which is why dividing the angles will not give a constant in the extension.*

### 4. What your results show

(a) Compare each pair of angles. Is  $\theta_2$  always bigger than  $\theta_1$ , always smaller, or does it vary?

*$\theta_2$  is always **smaller** than  $\theta_1$ . Going from air into perspex the ray always bends towards the normal.*

(b) Use the word **speed** to explain why.

*Light travels **slower** in perspex than in air. The side of the ray that reaches the perspex first slows down first, while the other side is still going quickly — so the ray swings round towards the normal. (This is the marching band crossing onto the long grass.)*

(c) What would happen to  $\theta_2$  if your block were made of **diamond** instead of perspex?

*( $n_{\text{perspex}} = 1.49$ ,  $n_{\text{diamond}} = 2.42$ )*

*$\theta_2$  would be even **smaller**. Diamond has a bigger  $n$ , so light is slower in it, so the ray bends more towards the normal.*

(d) Look at your  $0^\circ$  row. The ray came in **along the normal**. What did it do, and why?

*It carried straight on without bending:  $\theta_2 = 0^\circ$  as well. With  $\theta_1 = 0^\circ$  no part of the ray reaches the perspex before any other part, so there is nothing to swing it round. It still slows down — it just does not change direction.*

(e) The ray leaves the block through the **curved** face, and it does not bend there. Use the word **normal** to explain why not.

*Hint: the centre of the card is the centre of the circle, so the ray inside the block is travelling straight out along a **radius**.*

*The normal to a curved surface points straight out from the centre of the circle. The ray is travelling out along a radius, so it is already travelling **along the normal** — its angle of incidence at the curved face is  $0^\circ$ , and by part (d) a ray along the normal does not bend. That is the whole reason the block is this shape: it lets us study **one** boundary at a time.*

### Extension

For each row of your table, work out  $\theta_1 \div \theta_2$ .

Then work out  $\sin \theta_1 \div \sin \theta_2$ .

One of those two calculations gives roughly the **same answer every time**. Which one — and what is that number close to?

*$\sin \theta_1 \div \sin \theta_2$  is the constant one, and it comes out at about **1.5** — the index of refraction of perspex. Dividing the angles themselves does not give a constant, which is exactly what the bend in your graph was telling you. (This is Snell's law, and you have just measured it.)*